

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2046

G.M. Shrum 138 kV
December 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2046. The Interconnection Customer proposes a 64.8 MW battery energy storage facility interconnecting at the G.M. Shrum 138 kV substation in the Kentucky area. The request was received on March 30, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 551 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,864,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Louisiana 1 to Desoto Pleasant Run 138.0 kV
Contingency: Southaven Substation to Mistersky 138.0 kV L
Loading: 115.7% (Rating: 171 MVA)
DFAX: 9.2%
Pre-project loading: 112.3%
2. Facility: Wyco to S Hill 138.0 kV
Contingency: Station I to Bryan St 138.0 kV
Loading: 104.9% (Rating: 194 MVA)
DFAX: 10.6%
Pre-project loading: 101.4%
3. Facility: Keifer-Jackson Substation to Beaver Dam East 138.0 kV
Contingency: Tyranena to Lg&E; Canal 138.0 kV Line
Loading: 103.2% (Rating: 251 MVA)
DFAX: 8.1%
Pre-project loading: 101.2%
4. Facility: W Lebanon to Oakdale Power Sta 138.0 kV
Contingency: New Camp Substation to Hubbardstown 138.0 kV
Loading: 101.2% (Rating: 243 MVA)
DFAX: 13.1%
Pre-project loading: 97.7%
5. Facility: Louisiana 1 to New Camp 138.0 kV
Contingency: Melrose Electric to Wyco Substation 138.0 kV
Loading: 100.6% (Rating: 243 MVA)
DFAX: 10.2%
Pre-project loading: 97.9%

6. Facility: Hubbardstown to New Camp 138.0 kV
Contingency: Melrose Electric to Wyco 138.0 kV
Loading: 100.5% (Rating: 249 MVA)
DFAX: 10.1%
Pre-project loading: 97.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Louisiana 1 to Desoto Pleasant Run 138.0 kV	Rebuild line with higher-capacity conductor	\$16,028,000	6% (\$937,000)
Wyco to S Hill 138.0 kV	Reconductor line	\$4,427,000	5% (\$228,000)
Keifer-Jackson to Beaver Dam East 138.0 kV	Reconductor line	\$2,037,000	8% (\$166,000)
W Lebanon to Oakdale Power Sta 138.0 kV	Reconductor line	\$9,564,000	3% (\$284,000)
Louisiana 1 to New Camp 138.0 kV	Reconductor line	\$2,664,000	3% (\$73,000)
Hubbardstown to New Camp 138.0 kV	Reconductor line	\$6,458,000	3% (\$176,000)
	Total allocated to project		\$1,864,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.